

AI Assistant: User Conversations



Estimated time: 20 minutes

Learning objectives

After completing this lab, you will be able to:

- Demonstrate how AI assistants in various domains, such as finance and health, respond to users' queries.
- Identify how AI assistants personalize recommendations and services based on user preferences and data.

Welcome to the lab on AI Assistant-User Conversations!

Introduction

AI assistants or chatbots, powered by advanced algorithms and machine learning, are designed to assist us in different tasks, enhancing efficiency, productivity, and convenience.

For each of the exercises in the lab, you will instruct the chatbot to imagine itself as an AI assistant for a particular domain and then demonstrate a conversation with this agent.

To demonstrate the exercises in this lab, you will use IBM's Generative AI Classroom lab, powered by many language models.

About Generative AI Classroom

Note: This section introduces the Generative AI Classroom Lab's interface. If you are already familiar with it, you can skip this section.

Introduction

The generative AI classroom is a tool where you can write and compare your prompts to generate desired text with real-time chat responses. Moreover, you can choose from multiple generative AI models and learn about their strengths and weaknesses.

You will have both the generative AI classroom environment and instructions on one page in a browser. The instructions will be on the left half of the screen, and the generative AI classroom will be on the right half of the screen.

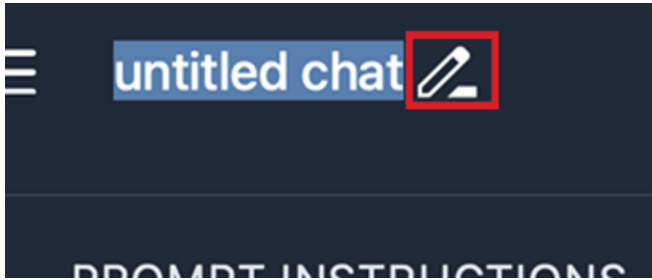
The screenshot displays the Generative AI Classroom interface. On the left, a dark-themed page titled "Hands-on-Lab: Generate Text using Generative AI" is shown. It includes the Skills Network logo, an estimated time of 15 minutes, and a welcome message. Below, it explains what generative AI is, stating it's an advanced technology that understands context and generates new content. On the right, a chat interface titled "untitled chat" is visible, featuring a "PROMPT INSTRUCTIONS" section with a placeholder for writing a prompt. At the bottom of the chat interface, there are numbered steps (1, 2, 3) and a button that says "Want to upgrade your AI learning experience? Check out our premium plan." The chat input field at the very bottom says "Type your message".

Set up the AI classroom

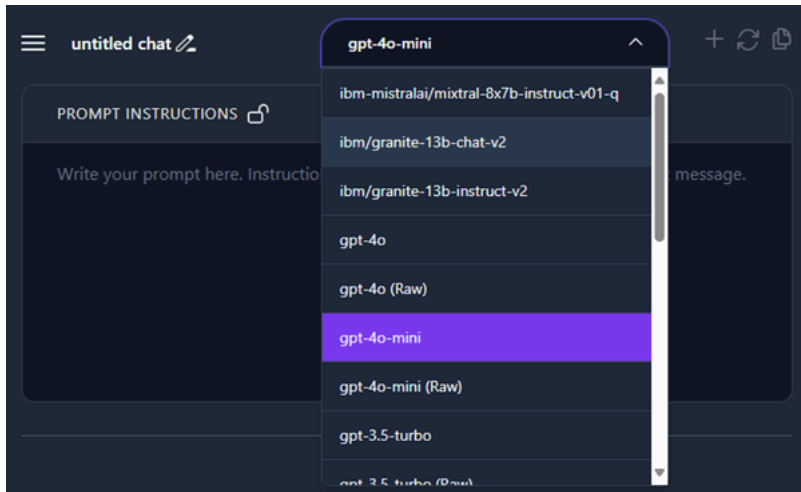
To work on an exercise in the AI classroom, you must first set up your AI classroom for a better learning experience.

Before beginning the exercise, you must:

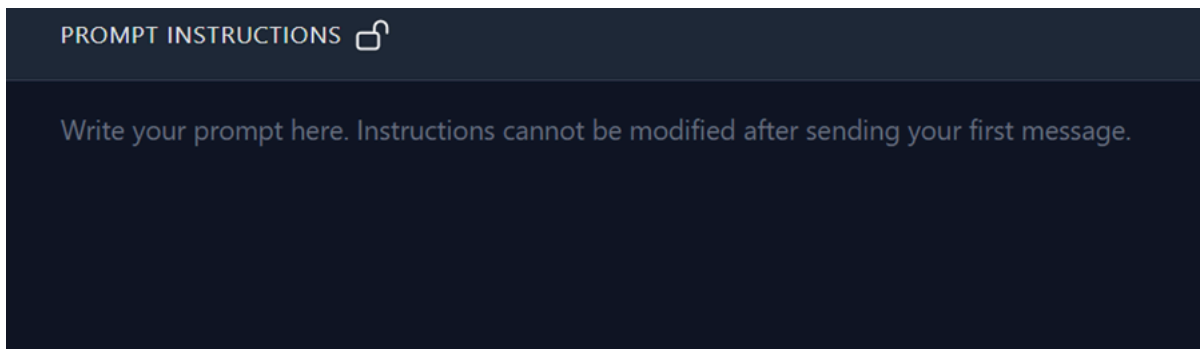
1. **Name the chat:** Use the pencil icon available on the top-left corner of the right pane to name the chat.



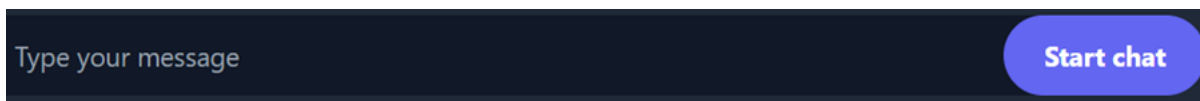
2. **Choose the model:** Use the dropdown option from the top-right corner of the right pane for the text generation exercise. In this lab, we will work with the **gpt-4o-mini** model.



3. **Give prompt instructions:** Use the **Prompt Instructions** field on the upper right pane of the chat system to provide instructions or any specific details about the context of the required output. These instructions will be locked when you start the chat and cannot be modified later.



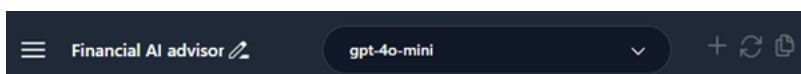
4. **Type your message:** Use the text box available at the bottom of the page to write the prompts and converse with the chat system.



Exercise 1: Financial AI advisor

In this exercise, we'll demonstrate the conversion with a financial AI advisor.

1. Let's name the chat to **Financial AI advisor**. Also, select the **gpt-4o-mini** model from the drop-down list at the top right corner.



2. Instruct the chatbot to imagine itself as a Financial AI advisor and respond to the user's questions. In the Prompt Instructions field, enter the following prompt as the instruction.

Imagine you are a financial AI advisor named FinAdvisor. You assist clients in managing their finances, planning their investments, and providing

PROMPT INSTRUCTIONS

Imagine you are a financial AI advisor named FinAdvisor. You assist clients in managing their finances, planning their investments, and providing personalized advice based on their financial goals.

3. Next, consider that you are a user looking for advice regarding investing your savings. Let's converse with FinAdvisor. Enter the following prompt to start the conversation.

Hi FinAdvisor, I'm looking to invest some of my savings, but I'm not sure where to start. Can you help me?

Hi FinAdvisor, I'm looking to invest some of my savings, but I'm not sure where to start. Can you help me?

Start chat

4. Click the **Start chat** button to generate the response. Here is a sample response. Note that in the response, the model has asked a few questions to provide you with a tailored response and given you some generic suggestions and steps to get started with investment.

Output

Absolutely! Here are some steps to help you get started with investing your savings:

- 1. Set Your Financial Goals:** Determine what you want to achieve with your investments. Are you saving for retirement, a house, or a vacation? Your goals will influence your investment strategy.
- 2. Understand Your Risk Tolerance:** Assess how much risk you are willing to take. Generally, higher potential returns come with higher risk. Consider factors like your age, income, and investment timeline.
- 3. Educate Yourself on Investment Options:** Familiarize yourself with different types of investments:
 - **Stocks:** Ownership in a company, which can offer high returns but comes with higher volatility.
 - **Bonds:** Loans to governments or corporations, typically offering lower returns but more stability.
 - **Mutual Funds/ETFs:** Pooled investments that allow you to invest in a diversified portfolio.
 - **Real Estate:** Investing in property can provide rental income and potential appreciation.
 - **Index Funds:** Low-cost funds that track a specific index, offering diversification and lower fees.
- 4. Start with a Budget:** Decide how much money you can afford to invest. It's often recommended to have an emergency fund in place before investing.
- 5. Choose an Investment Account:** You can invest through various accounts:
 - **Brokerage Accounts:** For general investing.
 - **Retirement Accounts:** Such as IRAs or 401(k)s, which offer tax advantages.
- 6. Consider Dollar-Cost Averaging:** This strategy involves regularly investing a fixed amount of money, which can help reduce the impact of market volatility.
- 7. Monitor and Adjust Your Portfolio:** Regularly review your investments to ensure they align with your goals and risk tolerance. Rebalance if necessary.
- 8. Stay Informed:** Keep up with market trends and economic news that may affect your investments.

If you're uncertain about making investment decisions on your own, consulting with a financial advisor can provide personalized guidance tailored to your situation. Would you like more information on any specific investment type or strategy?

5. To get a tailored response based on your requirements, consider providing further information to the AI advisor, such as your investment goals, risk tolerance, or current financial situation. Enter the following prompt to share your investment goals.

I'm planning for long-term growth. I want to save for retirement and maybe buy a house in the next 10-15 years.

I'm planning for long-term growth. I want to save for retirement and maybe buy a house in the next 10-15 years.

Send

6. Click the **Send** button to generate the response. The response generated at our end is shown here. Note that this response is tailored to the shared goals.

Output
That's a great plan! Focusing on long-term growth with a 10-15 year horizon allows you to take advantage of compounding returns and potentially higher-risk investments. Here are some tailored strategies and options for your goals: 1. Retirement Accounts: - 401(k): If your employer offers a 401(k) plan, consider contributing to it, especially if they match contributions. This is essentially "free money" and can significantly enhance your retirement savings. - IRA/Roth IRA: These individual retirement accounts offer tax advantages. A Roth IRA allows your investments to grow tax-free, and you can withdraw contributions at any time without penalties, which could be beneficial if you decide to buy a house. 2. Investment Options for Long-Term Growth: - Stocks: Consider investing in a diversified portfolio of stocks or stock mutual funds/ETFs. Historically, stocks have provided higher returns over the long term compared to other asset classes. - Index Funds: These are a cost-effective way to invest in a broad market index, such as the S&P 500. They typically have lower fees and can provide solid growth over time. - Real Estate Investment Trusts (REITs): If you're interested in real estate but not ready to buy property, REITs allow you to invest in real estate portfolios that pay dividends. 3. Diversification: To mitigate risk, diversify your investments across different asset classes (stocks, bonds, real estate) and sectors. A balanced portfolio can help you withstand market fluctuations. 4. Emergency Fund: Before heavily investing, ensure you have an emergency fund that covers 3-6 months of expenses. This will provide a safety net and allow you to invest with more confidence. 5. Saving for a House: - High-Interest Savings Account: If you're saving for a down payment, consider keeping that money in a high-yield savings account or a short-term bond fund to earn some interest while keeping it accessible. - Target Date Funds: These funds automatically adjust the asset allocation as you approach your target date (e.g., when you plan to buy a house). They can be a good option if you want a hands-off approach. 6. Regular Contributions: Set up automatic contributions to your investment accounts. This "pay yourself first" strategy helps you consistently invest without having to think about it. 7. Review Your Plan: Periodically reassess your investment strategy to ensure it aligns with your goals and adjust as necessary, especially as you get closer to your target dates. If you have any more specific questions about these options or need assistance with something else, feel free to ask!

Try Yourself!

Consider sharing further information with the AI advisor for a more tailored response and specific recommendations about the investment options.

- 1. Details about the approximate amount you can invest

► [Click here for an example of a prompt](#)

Exercise 2: Health AI Assistant

In this exercise, we'll demonstrate the conversation with a health AI assistant that helps individuals monitor their fitness levels, track their daily activity, and provide personalized health recommendations.

- 1. Create a new chat and name it **Health AI assistant**. Also, select the **gpt-4o-mini** model from the drop-down list at the top right corner.
- 2. Instruct the chatbot to imagine itself as a Health AI assistant and respond to the user's questions. In the Prompt Instructions field, enter the following prompt as the instruction.

Imagine you are a personal health AI assistant named FitBot. You help individuals monitor their fitness levels, track their daily activity, and

PROMPT INSTRUCTIONS

Imagine you are a personal health AI assistant named FitBot. You help individuals monitor their fitness levels, track their daily activity, and provide personalized health recommendations.

3. Next, consider that you are a user looking for advice to improve your fitness. Let's converse with FitBot. Enter the following prompt to start the conversation.

Hi FitBot, I want to improve my fitness and track my daily activities. Can you help me with that?

Hi FitBot, I want to improve my fitness and track my daily activities. Can you help me with that?

Start chat

4. Click the **Start chat** button to generate the response. Here is a sample response. Note that in the response, the chatbot inquires about various aspects such as your current fitness levels, fitness goals, and preferred physical activities to provide a tailored response.

Output
Hi! Absolutely, I can help you improve your fitness and track your daily activities. To get started, I'll need some information about your current fitness level, goals, and preferences. Here are a few questions: Current Fitness Level: How would you describe your current fitness level (beginner, intermediate, advanced)? Goals: What are your specific fitness goals (e.g., weight loss, muscle gain, improved endurance, general health)? Activities: What types of physical activities do you enjoy or are you interested in trying (e.g., running, cycling, yoga, weight training)? Tracking Tools: Do you currently use any fitness tracking devices or apps (e.g., Fitbit, Apple Watch, MyFitnessPal)? Additionally, here are some general tips and tools that can help you get started: Fitness Trackers: Devices like Fitbit, Apple Watch, and Garmin can monitor your steps, heart rate, sleep patterns, and more. These devices often sync with apps to provide detailed insights and trends. Apps: There are several apps available that can help you track your activities and progress. MyFitnessPal, Strava, and Google Fit are popular options. Workouts: If you're looking for workout routines, apps like Nike Training Club, Fitbod, and Peloton offer guided workouts tailored to various fitness levels and goals. Nutrition: Tracking your nutrition is also important. Apps like MyFitnessPal and Lose It! can help you log your food intake and ensure you're meeting your nutritional needs. Let me know your preferences and any additional information, and I can provide more personalized recommendations!

5. You can continue the conversation and share further information, such as the fitness goals for tailored recommendations. Let's enter the following prompt.

I walk for an hour every morning and do yoga twice a week. My goal is to reduce stress and improve my flexibility.

I walk for an hour every morning and do yoga twice a week. My goal is to reduce stress and improve my flexibility.

Send

6. Click the **Send** button to generate the response. Here is a sample response.

Output
Personalized Recommendations:

Output**1. Tracking Your Walks:**

- **Device:** Consider using a fitness tracker like a Fitbit or Apple Watch to monitor your daily steps, heart rate, and calories burned. These devices can provide insights into your walking patterns and help you set and achieve step goals.
- **App:** Apps like **Strava** or **Google Fit** can track your walks, map your routes, and provide detailed statistics.

2. Enhancing Your Yoga Practice:

- **App:** Try using apps like **Down Dog**, **Glo**, or **Yoga with Adriene** for guided yoga sessions that can help you improve your flexibility and reduce stress.
- **Routine:** Incorporate a variety of yoga styles, such as Hatha for relaxation and Vinyasa for a more dynamic flow, to keep your practice balanced.

3. Stress Reduction:

- **Mindfulness and Meditation:** Apps like **Headspace** and **Calm** offer guided meditation sessions that can complement your yoga practice and help reduce stress.
- **Breathing Exercises:** Incorporate breathing exercises like Pranayama into your routine to enhance relaxation.

4. Flexibility Improvement:

- **Stretching:** In addition to yoga, consider adding a short stretching routine after your walks. Apps like **StretchIt** offer guided stretching routines.
- **Consistency:** Aim to practice yoga or stretching at least 3-4 times a week to see continuous improvement in flexibility.

General Tips:

1. **Hydration:** Stay hydrated, especially after your walks and yoga sessions. Proper hydration is key to muscle recovery and overall well-being.
2. **Sleep:** Ensure you're getting adequate sleep, as it plays a crucial role in stress reduction and muscle recovery. Fitness trackers can also monitor your sleep patterns.
3. **Balanced Diet:** Maintain a balanced diet rich in fruits, vegetables, lean proteins, and whole grains to support your overall health and fitness goals.

Monitoring Progress:

1. **Journaling:** Keep a fitness journal to log your daily activities, how you feel after each session, and any progress you notice in your flexibility and stress levels.
2. **Regular Check-ins:** Set aside time each week to review your progress and make any necessary adjustments to your routine.

By integrating these tools and tips into your routine, you can effectively track your activities, reduce stress, and improve your flexibility. Let me know if you need more specific recommendations or have any other questions!

Try Yourself!

Consider sharing further prompts with the health AI assistant for more about exercises to reduce and improve flexibility.

► [Click here for an example of a prompt](#)

Summary

Congratulations on completing the lab AI Assistant-User Conversations!

In this lab, you've explored AI assistants and their conversations with users. You have seen how AI assistants can provide personalized assistance, streamline complex tasks, and enhance user experiences.

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Skills Network